

FAIZAH NAQVI



(732) 567-5976



faizahnaqvi@gmail.com



<https://ffnaqvi.github.io/>

EDUCATION

Brown University

Providence, RI

Bachelor & Master of Science in Computer Science | **GPA:** 3.96/4.0, Magna Cum Laude (Top 20%)

May 2026

- Relevant coursework:** Object-Oriented Programming, Deep Learning/AI, Computer Systems, High Performance Network Systems, Systems Security, Database Management, Applied Cryptography.
- Activities:** Women in Computer Science Club (Mentor), Community Coordinator (Upper Division).

SKILLS

Programming Languages: Python, C/C++, SQL, Java, Rust, Go, C#, JavaScript.

Certifications: AWS Cloud Practitioner, JNCIA-Junos.

Tools & Frameworks: Linux, Git, Docker, Grafana, REST APIs, React, .NET.

PROFESSIONAL EXPERIENCE

Juniper Networks

Sunnyvale, CA

Software Engineer Intern, Test Engineering

May 2025 – August 2025

- Engineered and tested virtual router topologies, tracing distributed system issues to their root cause **to resolve bugs impacting over 100 features** across multiple Junos releases.
- Owned end-to-end validation of Juniper's software for industry compliance (gNMI and OpenConfig) **across ~7,000 features** in multiple Junos releases.
- Designed and implemented containerized Go applications for Juniper routers, including a gRIBI proxy that **enhanced the reliability of programmable routing pipelines**.

Consolidated Edison

New York, NY

Information Technology Intern, Enterprise Architecture

May 2024 – August 2024

- Assessed application and infrastructure architectures **to identify security gaps across APIs and services**, recommending remediation strategies aligned with runtime protection and continuous compliance goals.
- Designed a **company-wide SDLC framework to mitigate security vulnerabilities** throughout the software development life cycle, focusing on the OWASP Top Ten, authentication, and access control.

Brown University Department of Computer Science

Providence, RI

Undergraduate Teaching Assistant

May 2023 – May 2026

- Reviewed code, conducted technical and conceptual office hours, and evaluated projects on **core systems and algorithm topics** alongside ~20 colleagues.
- Collaborated directly with Prof. Van Dam to research, craft, and deliver biweekly mini-lectures on socially responsible computing to **~400 students**.

RESEARCH

Brown University Systems Research Group

Providence, RI

Undergraduate Researcher | SQL, C, Rust

September 2025 – May 2026

- Design and develop a novel system that translates SQL-like queries into highly optimized, safe programs that **run directly on the Linux kernel (eBPF)**.
- Conduct systems research under Professor Malte Schwarzkopf as part of the Efficient and Trustworthy Operating Systems Group; work with postdoc researchers **on kernel-level observability**.

AI Robotics Ethics Society @ Brown

Providence, RI

Undergraduate Researcher | Python

September 2023 – May 2025

- Co-author** of Crossing the Principle-Practice Gap in AI Ethics with Ethical Problem-Solving. Developed a framework that evaluates AI model risk for software developers.
- Led team of five undergraduates** to conduct a study measuring the impact of technology ethics education within the Computer Science Department.

SELECTED PROJECTS

Low-Latency Network System | Rust

March 2025

- Implemented and analyzed a high-performance multi-threaded TCP client-server application with I/O ring optimizations for open-loop request generation, **reducing latency by 44%** and maximizing system throughput.

American Community Survey (ACS) Data Connector | JavaScript, React, Java

April 2024

- Built an accessible data visualization web app consuming U.S. Census ACS REST APIs, implementing full keyboard navigation and screen-reader support to meet WCAG accessibility standards.